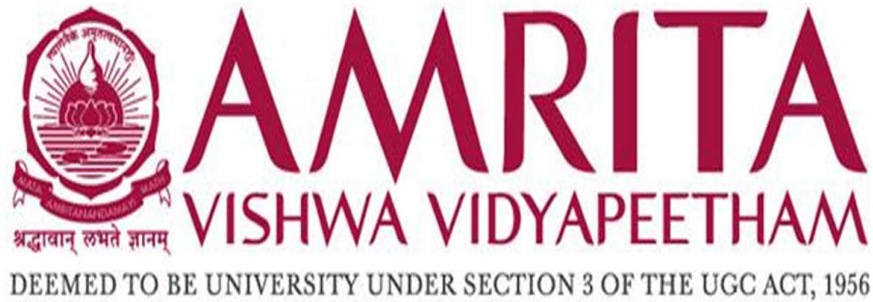




AMRITA SCHOOL OF ENGINEERING
BENGALURU

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

23ECE184- IoT LAB

**A Report on
IoT System Design Project-
Smart Street Lighting System Using Arduino UNO, LDR Sensor,
PIR Sensor, and LED Array**

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1. Title

Smart Street Lighting System Using Arduino UNO, LDR Sensor, PIR Sensor Array, and MIT App Inventor

2. Objectives

This project aims to build a smart, energy-saving street lighting system that turns lights on only when they are actually needed — reducing electricity waste. Below are the main objectives:

- Automatically detect day and night using an LDR sensor, keeping all lights OFF during the day and switching them to a dim standby at night.
- Detect movement using PIR sensors across three zones. When motion is detected, only the relevant zone's light brightens fully.
- After motion stops, hold the light bright for 5 seconds, then gradually dim it back over 80 smooth steps instead of a sudden switch-off.
- Allow BBMP and BMTC officials to manually control any zone light through an Android app (MIT App Inventor) over Bluetooth.
- Simulate and validate the full circuit in Tinkercad before physical assembly.

3. Components Required

The table below lists all hardware and software used in this project:

Component	Specification	Qty	Purpose
Arduino UNO	ATmega328P, 16 MHz	1	Main controller
PIR Sensor	HC-SR501, 5V	3	Motion detection (one per zone)
LDR Sensor	GL5528, 10 kΩ pull-down	1	Detects day vs. night
LED (Red, 5 mm)	2.2V forward, ~20 mA	4	Simulates street lights
Resistor 220 Ω	¼W carbon film	4	Current limiting for LEDs
Resistor 10 kΩ	¼W carbon film	1	LDR voltage divider
Breadboard	830 tie-points	1	Prototyping
Jumper Wires	Male-to-Male	~30	Connections
USB Cable (Type B)	5V, 500 mA	1	Power and programming
Android Phone	Android 5.0+, Bluetooth	1	Runs the BBMP control app

Software Tools

- Arduino IDE — for writing and uploading code to the Arduino UNO
- Tinkercad — for designing and testing the circuit in simulation

- MIT App Inventor — for building the BBMP Android control app
- GitHub — for storing and sharing the source code

4. Block Diagram

The diagram below shows how all the parts of the system connect. The Arduino UNO is the brain — it reads the LDR and three PIR sensors as inputs, and controls three LED lights as outputs. The BBMP app connects over Bluetooth and lets officials override any zone manually.

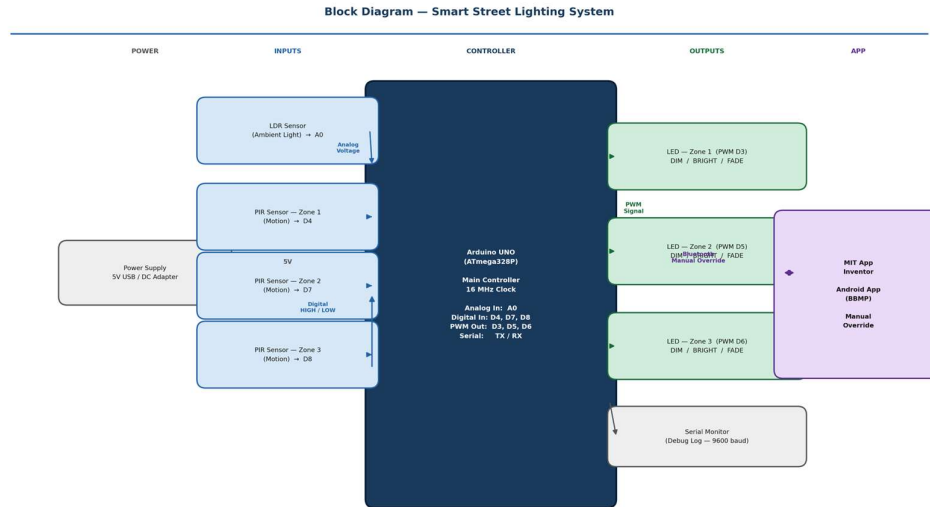


Figure 1: Block Diagram — Power → LDR + PIR sensors → Arduino UNO → LED zones + MIT App override via Bluetooth

5. Circuit Diagram

The circuit was built and tested in Tinkercad. The pin connections are shown below:

Component	Arduino Pin	Notes
LDR Sensor	A0 (Analog In)	Reads light level (0–1023)
PIR — Zone 1	D4 (Digital In)	Motion detected = HIGH
PIR — Zone 2	D7 (Digital In)	Motion detected = HIGH
PIR — Zone 3	D8 (Digital In)	Motion detected = HIGH
LED — Zone 1	D3 (PWM Out)	Brightness via analogWrite()
LED — Zone 2	D5 (PWM Out)	Brightness via analogWrite()
LED — Zone 3	D6 (PWM Out)	Brightness via analogWrite()

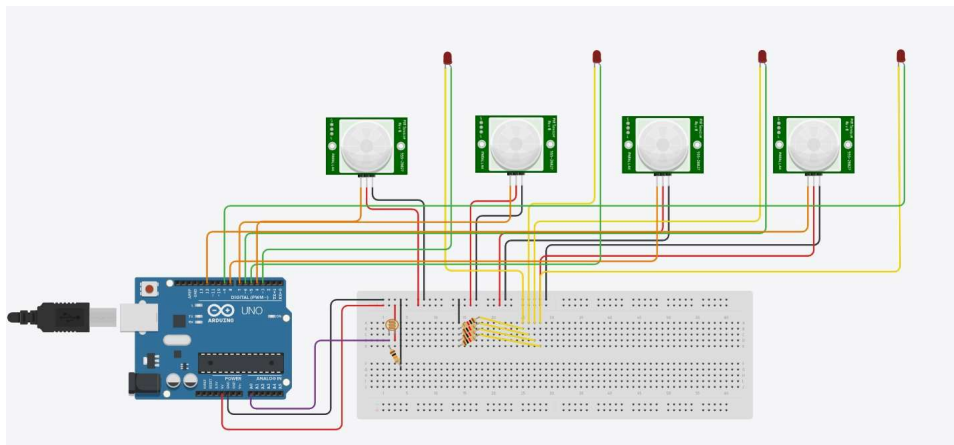


Figure 2: Circuit Diagram — Tinkercad simulation showing all three zones wired to the Arduino UNO with LDR on A0 and PWM LEDs on D3, D5, D6

6. Working Prototype

How It Works

When the system powers up, the Arduino continuously reads the LDR. If the reading is above 700, it considers it daytime and keeps all LEDs off. Once the reading drops below 500 (dark enough), it switches to Night Mode and dims all LEDs to a low standby brightness (PWM = 40 out of 255).

If a PIR sensor picks up movement in any zone, that zone's LED immediately jumps to full brightness (PWM = 255). The light stays fully on for 5 seconds after the last detected movement, then fades back smoothly over 80 steps. If motion is detected again during the fade, the light snaps back to full brightness right away.

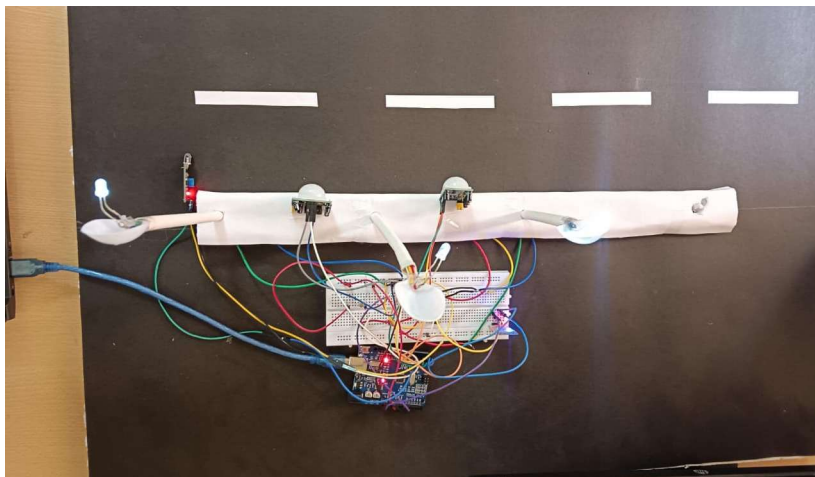


Figure 3: Physical prototype — Arduino UNO, breadboard, PIR sensors, LDR, and LED array

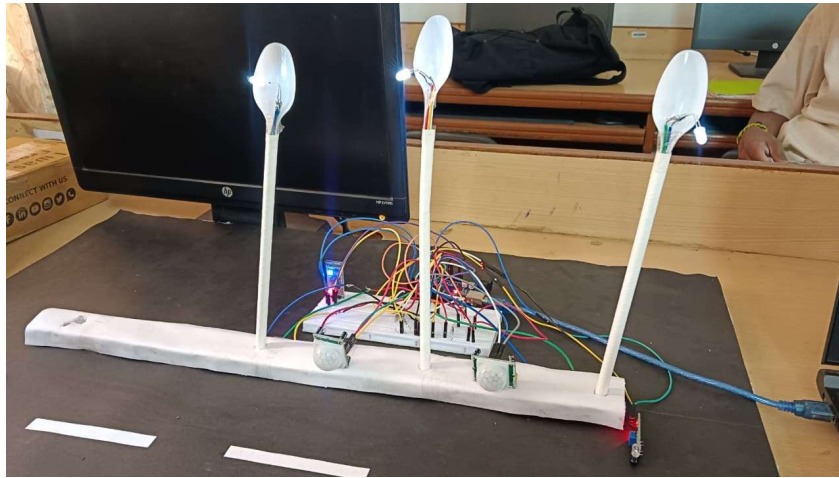


Figure 4: Prototype in Night Mode — motion detected in Zone 1, LED at full brightness (PWM 255); other zones at standby dim (PWM 40)

7. MIT App Inventor — Screenshots

A dedicated Android app titled "BBMP — Smart Street Lights" was created using MIT App Inventor. It lets BBMP and BMTc officials manually turn any zone ON or OFF, overriding the automatic control. This is useful during road works, events, or emergencies.

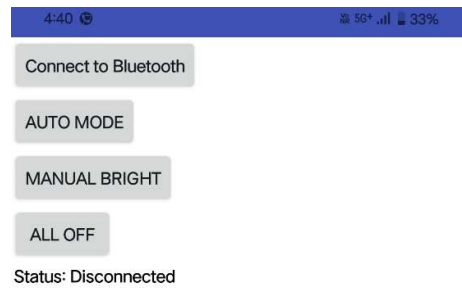


Figure 5: MIT App home screen — BBMP logo, title, and zone control dashboard in BBMP brand indigo/purple

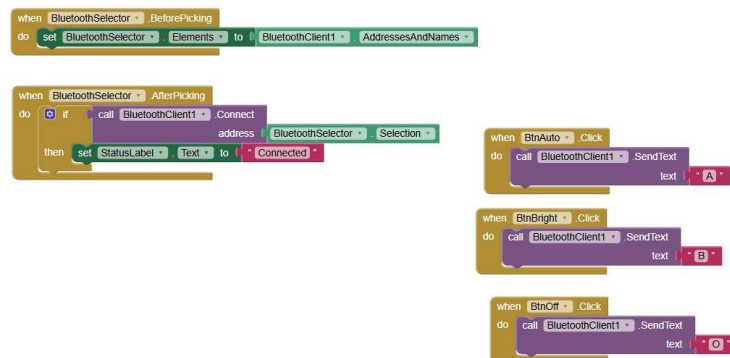


Figure 6: MIT App zone controls — individual ON/OFF buttons for Zone 1, Zone 2, and Zone 3

The app sends a Bluetooth command string (e.g. "Z1ON" or "Z2OFF") to the Arduino, which reads it and overrides the PIR logic for that zone until the command is cleared.

8.2 MIT App Inventor — Flow Chart

This flowchart shows the MIT App logic — from opening the app and connecting via Bluetooth, to sending zone commands, updating the UI, and exiting.

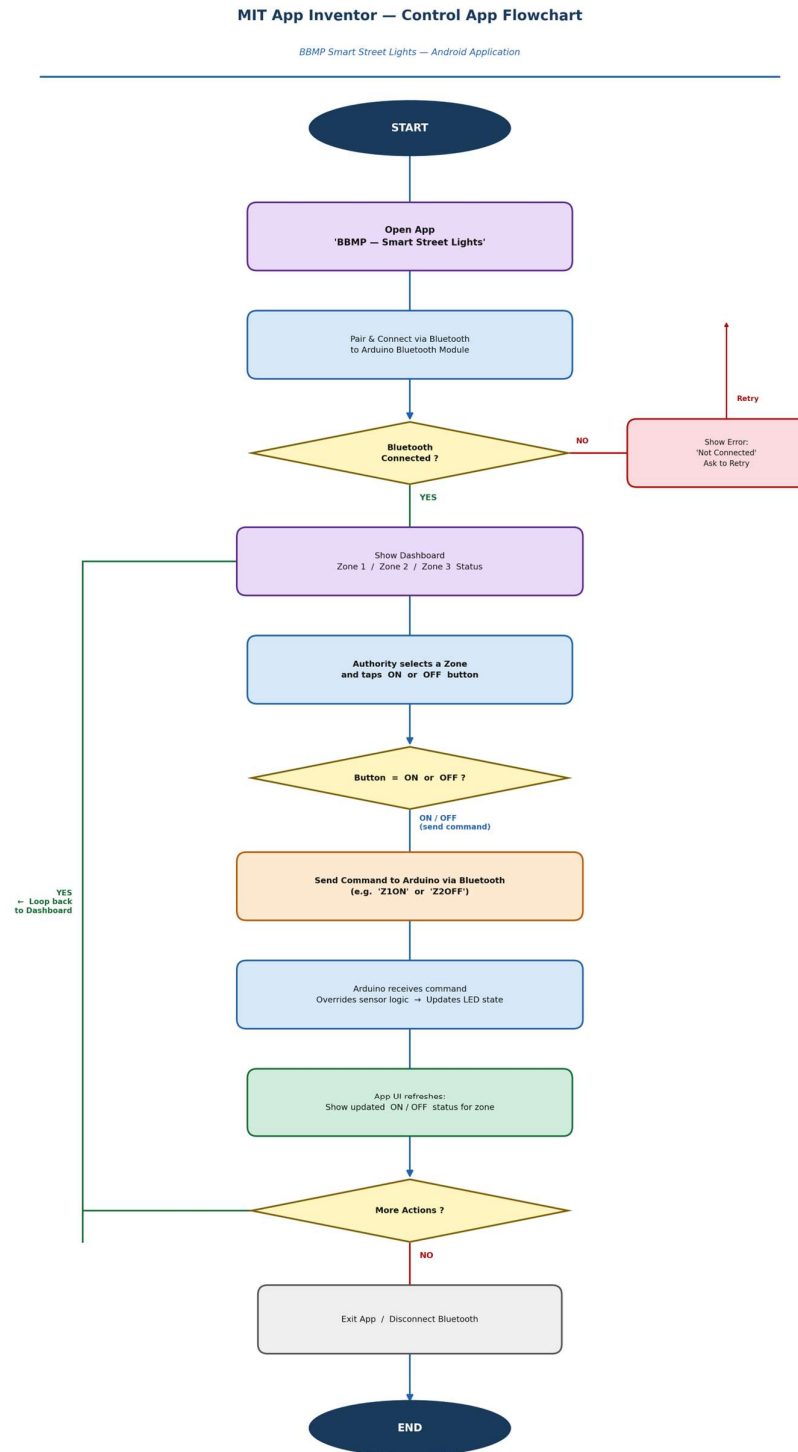


Figure 8: MIT App flowchart — Open App → Bluetooth connect → Dashboard → Tap zone button → Send command → Arduino updates LED → UI refreshes → Loop / Exit

9. GitHub Link

The complete Arduino source code is uploaded to GitHub. The repository includes the main .ino file, a README with pin mapping, and the Tinkercad simulation link.

10. References

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- Arduino Official Documentation — <https://docs.arduino.cc>
 - HC-SR501 PIR Sensor Datasheet — <https://components101.com/sensors/hc-sr501-pir-sensor>
 - MIT App Inventor Documentation — <https://appinventor.mit.edu/explore/ai2>
 - Tinkercad Circuit Simulator — <https://www.tinkercad.com>
 - BBMP Official Website — <https://bbmp.gov.in>
 - "Smart Street Lighting using IoT" — IEEE Xplore, various authors, 2022–2024
 - Arduino PWM Tutorial — <https://docs.arduino.cc/tutorials/generic/secrets-of-arduino-pwm>